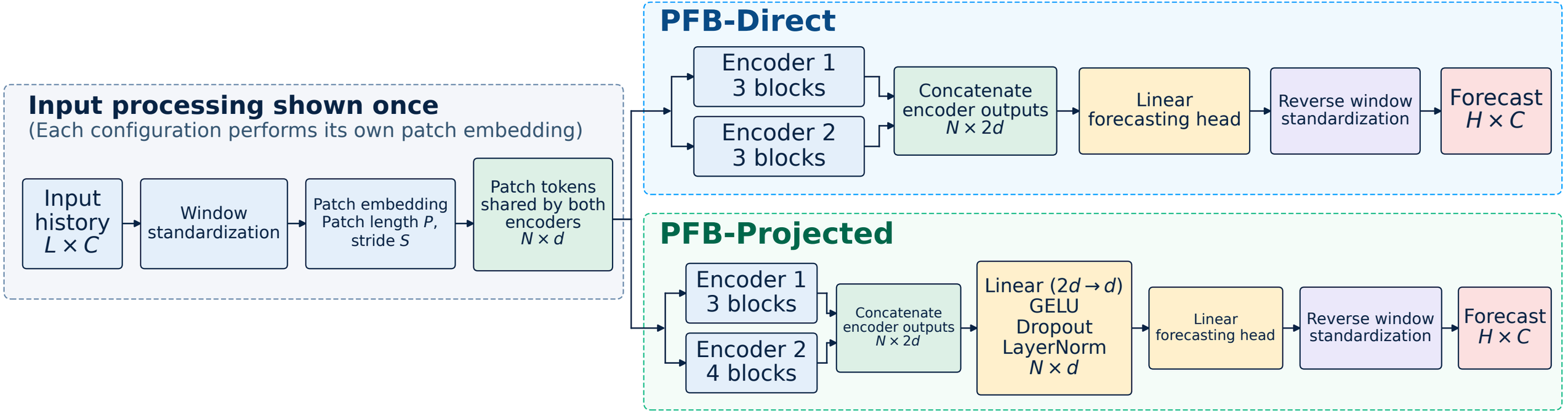PFB-Direct and PFB-Projected